

Assignment 4 CHE 321

Q5-12, P5-1 (b), (d), + (e), P5-5, P5-8, P5-11

Q5-12

Q5-12_A Go to the five LearnChemE screencasts link for Chapter 5
(<http://www.umich.edu/~elements/6e/05chap/learn-cheme-videos.html>).

1. In the screencast of the PBR with pressure drop, is there a problem substituting for the volumetric flow rate when there is a change in the total number of moles, that is, $\delta \neq 0$?
2. View one of the other screencasts 5- to 6-minute video tutorials and list two of the most important points.
3. In the isothermal plug reactor screencast, how does the inert impact the reaction process?

- 1) Yes, there's a problem substituting volumetric flow rate when there is Δn the total # of moles, the total molar flow rate Δ 's w/ conversion + pressure drop & it Δ 's along reactor. So you have to take into account both total moles + Δ pressure using ideal gas law rather than assuming constant volumetric flow.
- 2) - For isothermal PFR is that the design eqn is written in differential form as $\frac{dF_A}{dV} = -r_A$ Specifically because rxn rate often is dependent on X .
- For a gas rxn volumetric flow rate can Δ w/ conversion if total # of moles Δ 's. Molar concentration needed to be expressed in terms of molar flow rate, pressure & Temp.
- 3) - An inert dilutes reactants, lowering their concentration & partial pressure, which reduces reaction rate. It can also increase total molar flow rate increasing volumetric flow rate & lowering overall X .

P5-1 b)

(I) After analyzing the parameters in the Ergun eqn. it was determined that A_c (Area of catalyst bed) has the greatest impact on pressure drop. By adjusting A_c , the C_1

term in the beta expression which has influence in the pressure vs z eqn

(II) - Decreasing the pressure drop, the area of the catalyst needs the utmost care. When changing viscosity it has the least effect on pressure drop & can be removed.

- The other parameters that effect the pressure drop are mass flow rate & diameter of particle

P5-1(c)

I) - conversion profile by varying parameters K , C_{A0} + V_0 whereas pressure drop depends upon parameter of increasing K + C_{A0} also increases the conversion for a given weight & decreases for the case of Volumetric flow rate

- Increasing α increases pressure drop.

$$\frac{dP}{dW} = \frac{-\alpha}{2/P}$$

$$\frac{dV}{dW} = -\frac{f_0}{F_{A0}}$$

(II) All 3 concentration curves decrease as we approach the catalyst weight of 55 kg. As we towards the end of reactor X becomes constant + pressure drops rapidly. C_1 & C_2 are directly dependent on conversion so they both decrease as W

(III) - The entering volumetric flow rate has the greatest effect on conversion whereas pressure drop is function of α

P5-1(e)

I) X increases + f decreases w/ an increase in K' for the same W , while P remains unchanged, & vice versa. This is expected because as K' increases, the rate of reaction increases & conversion increases

(II) - α & F_{A0} increase causes X & P to decrease for the same W , while f increases

- X down reactor increases, it causes more product to be formed & lesser amounts of reactants are present. $\alpha < 0$ the volumetric flow rate decreases with an increase in conversion

(III)

I can conclude that conversion is improved by higher reaction rates

but performance can be impacted significantly by pressure drop

α & K must be considered for optimal reactor design so reaction kinetics must be considered & pressure loss at the same time

IV) - In regards to future chapters what if the system or reaction is not isothermal?

- What if we had inert gas in our rxn?

PS-5

2A → B

$$r = -kC^2$$

$$X = 0.5$$

$$F = 500 \frac{\text{mol}}{\text{h}}$$

$$C_{A0} = 0.2 \frac{\text{mol}}{\text{dm}^3}$$

$$\frac{dC}{dt} = -kC^2$$

$$\int_{C_0}^{C_0/2} \frac{dC}{C^2} = -k \int_0^t dt$$

$$\frac{1}{C_{A0}(1-X)t} - \frac{1}{C_{A0}t} = k$$

$$kC_0 = 1$$

$$k = 5 \frac{\text{dm}^3}{\text{mol h}}$$

$$-r_A = 5 \frac{\text{dm}^3}{\text{mol h}} \left(0.2 \frac{\text{mol}}{\text{dm}^3} (0.5) \right)^2$$

CSTR

$$V = \frac{F_1 X}{kC^2(1-X)^2} = \frac{500 \times 0.5}{5 \times 0.2^2 (1-0.5)^2} = 5000 \text{ dm}^3$$

PS-8

$$\frac{dV}{dV} = \frac{F_B}{F_{A0}}$$

$$-r_A = kC_A$$

$$C_A = C_{A0}(1-X)$$

$$\frac{dX}{dV} = \frac{kC_{A0}}{F_{A0}} (1-X) = \frac{k}{V_0} (1-X)$$

$$\ln \frac{1}{1-X} = kT$$

$$X = 0.632$$

$$\ln 2.73 = kT$$

$$kT = 1$$

CSTR Added upstream of PFR

$$V = \frac{F_{A0} X}{kC_{A0}(1-X)}$$

$$TK = \frac{X}{1-X}$$

$$\frac{KV}{V_0} = \frac{x}{1-x} - 1$$

$$x = 0.5$$

$$x_1 = 0.5$$

finding x_2

$$V = F_{A0} \int_{x_1}^{x_2} \frac{1}{-r_A} dx = V_0 C_{A0} \int_{x_1}^{x_2} \frac{1}{K C_{A0} (1-x)} dx$$

$$TK = \int_{x_1}^{x_2} \frac{dx}{1-x}$$

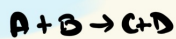
$$\ln \frac{1}{1-x_2} - \ln \frac{1}{1-x_1}$$

$$1 = \ln \frac{1}{1-x_2} - \ln \frac{1}{1-0.5}$$

$$1 + \ln 2 = 1.69 = \ln \frac{1}{1-x_2}$$

$$x_2 = 0.82$$

P5-11



$$F_{A0} \frac{dx}{dW} = -r_A'$$

$$-r_A' = k C_A C_B$$

$$C_A = \frac{F_A}{V} = \frac{F_{A0}(1-x)}{V} = C_{A0}(1-x)P$$

$$P = \frac{P_0}{F_0} = (1-\alpha W)^2$$

$$C_A = C_{A0}(1-x)(1-\alpha W)^2$$

$$-r_A' = k C_{A0}^2 (1-x)^2 (1-\alpha W)$$

$$\frac{dx}{dW} = \frac{k C_{A0}^2 (1-x)^2 (1-\alpha W)}{F_{A0}}$$

$$\int_0^x \frac{dx}{(1-x)^2} = \frac{k C_{A0}^2}{F_{A0}} \int_0^W (1-\alpha W) dW$$

$$\int_0^x \frac{1}{(1-x)^2} dx = \frac{1}{1-x} - 1 = \frac{x}{1-x}$$

$$\int_0^W (1-\alpha W) dW = W - \frac{\alpha W^2}{2} = W \left(1 - \frac{\alpha W}{2}\right)$$

$$\frac{x}{1-x} \left(\frac{F_{A0}}{K_c^2 a_0} \right) = W \left(1 - \frac{\alpha W}{2} \right)$$

$$\alpha \cdot W = 1 - \left(\frac{p}{p_0} \right)^2 = 0.99$$

$$X = 0.8 \quad F_{A0} = 10 \frac{\text{mol}}{\text{min}} + C_{A0} = 0.4 \frac{\text{mol}}{\text{dm}^3}$$

$$K \cdot W = \frac{\frac{0.8}{1-0.8} \times \frac{10}{0.4^2}}{\frac{1-0.99}{2}} = 495.04$$

Increasing particle size: $d_p = 2 \cdot d_1$

$$\alpha_2 = 0.5 \alpha_1$$

$$K \cdot W = 495.04$$

$$\alpha_2 \cdot W = 0.495$$

$$\frac{x}{1-x} \left(\frac{F_{A0}}{K_c^2 a_0} \right) = W \left(1 - \frac{\alpha W}{2} \right)$$

$$X = 0.886 = 88.6\%$$